

Page 1 - Current lamps

Lamp count: 40 halogen spots.

Power per halogen lamp: 60 W.

Power per replacement LED lamp: 8 W.

Typical LED lifetime: 25,000 hours. Halogen lifetime: 2,000 hours.

Replacement halogen lamp price: EUR 4 each.

Page 2 - Usage

Opening days per year: 220 (closed weekends and holidays).

Lights on per opening day: 11 hours.

Measured occupancy: room is empty about 35% of the time lights are on.

Most empty periods: 08:00-09:30 and 17:00-19:00.

Page 3 - Prices and quotes

Electricity price (business contract, 2026): EUR 0.28 per kWh incl. taxes.

Option A - basic LED lamp: EUR 18 per lamp, no installation cost (swap-in).

Option B - smart LED lamp: EUR 65 per lamp.

Option B - motion sensors + installation: EUR 400 one-off.

Student board budget for this project: EUR 2,500.